Haven AI

(Real Estate Support Triage Agent)

Manufacturing Operational Insights Report

Course Name: GEN AI

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Problem Statement & Objectives

1. Problem Statement

The primary objective of this project is to develop an AI-powered educational system for the real estate domain that can generate interactive quizzes and provide personalized explanations to users. Traditional online quiz systems rely on fixed questions and exact answer matching, which limits personalization and deeper learning.

In the real estate domain, users often need conceptual understanding of property types, pricing, legal procedures, and market terminology. Existing systems do not provide intelligent tutoring or contextual feedback. Therefore, there is a need for a Generative AI-based solution that can dynamically generate domain-specific questions and evaluate responses using natural language understanding.

To address this challenge, the proposed system uses a Large Language Model (LLM) through Groq API to generate questions, assess user answers, and provide detailed explanations. This creates a personalized learning environment for users interested in real estate knowledge.

2. Project Objectives

The key objectives of this project are:

- Develop an **interactive educational bot**
- Use **LLM APIs (Groq / OpenAI compatible)**
- Generate **dynamic quiz questions**
- Evaluate user answers intelligently
- Provide detailed AI-generated explanations
- Maintain user quiz sessions and scores
- Improve learning using personalized feedback

The system focuses on making learning more interactive and domain-specific.

The main objective of this project is to build an intelligent real estate learning assistant that can conduct assessments automatically using Generative AI. The system aims to replace static educational methods with a dynamic learning platform that adapts to the user's responses.

3. Scope of the Project

The scope of this project includes the development of an AI-driven quiz and tutor system specifically for the real estate domain. The system supports dynamic question generation, answer evaluation, and contextual feedback generation using a Large Language Model.

The project includes a user interface for quiz interaction, backend APIs for processing requests, and local storage for maintaining quiz records. It also supports future expansion into multilingual tutoring, voice interaction, and cloud deployment.

However, the project does not currently include real-time property listing integration, advanced analytics, or enterprise-level deployment. It focuses primarily on educational assistance within the real estate domain.

Proposed Solution

The proposed solution is a fully automated and scalable system that integrates AI-based query handling with DevOps-driven infrastructure provisioning. The system uses Terraform or Ansible to define and provision infrastructure, ensuring consistent and repeatable setup.

The application is containerized using Docker, which packages the backend, dependencies, and configurations into a single deployable unit. Kubernetes is used to manage these containers by handling deployment, scaling, and load balancing automatically.

The backend is built using FastAPI, while MongoDB is used for data storage. AI models process user queries and generate structured responses. This combination of technologies ensures high performance, scalability, and ease of deployment.

1. Key Features

1. Dynamic Quiz Generation

- Generates unique real estate questions automatically
- Avoids repeated static questionnaires

2. Intelligent Answer Evaluation

- Understands semantic meaning of user responses
- Scores answers accurately using LLM

3. Personalized Explanations

- Explains incorrect answers in detail
- Helps users improve understanding

4. User-Friendly Interaction

- Simple chatbot-style quiz interface
- Easy navigation for users

5. Scalable Architecture

- Supports future upgrades
- Easy integration with cloud deployment

2. Overall Architecture / Workflow

The workflow consists of multiple layers.

The user interacts with the quiz interface.

The request is sent to the backend.

The backend communicates with the LLM API.

The LLM generates:

- question
- answer evaluation
- explanation

The result is returned to the frontend.

The complete workflow is:

User → Backend → LLM API → Evaluation → Response

This ensures real-time interaction and personalized educational support

3. Tools & Technologies Used

Python

- Core programming language
- Used for backend development

FastAPI

- Builds RESTful APIs
- Handles user requests efficiently

Groq API

- Provides LLM inference
- Generates and evaluates quiz content

Local Storage

- Stores quiz data
- Maintains user session records

Frontend

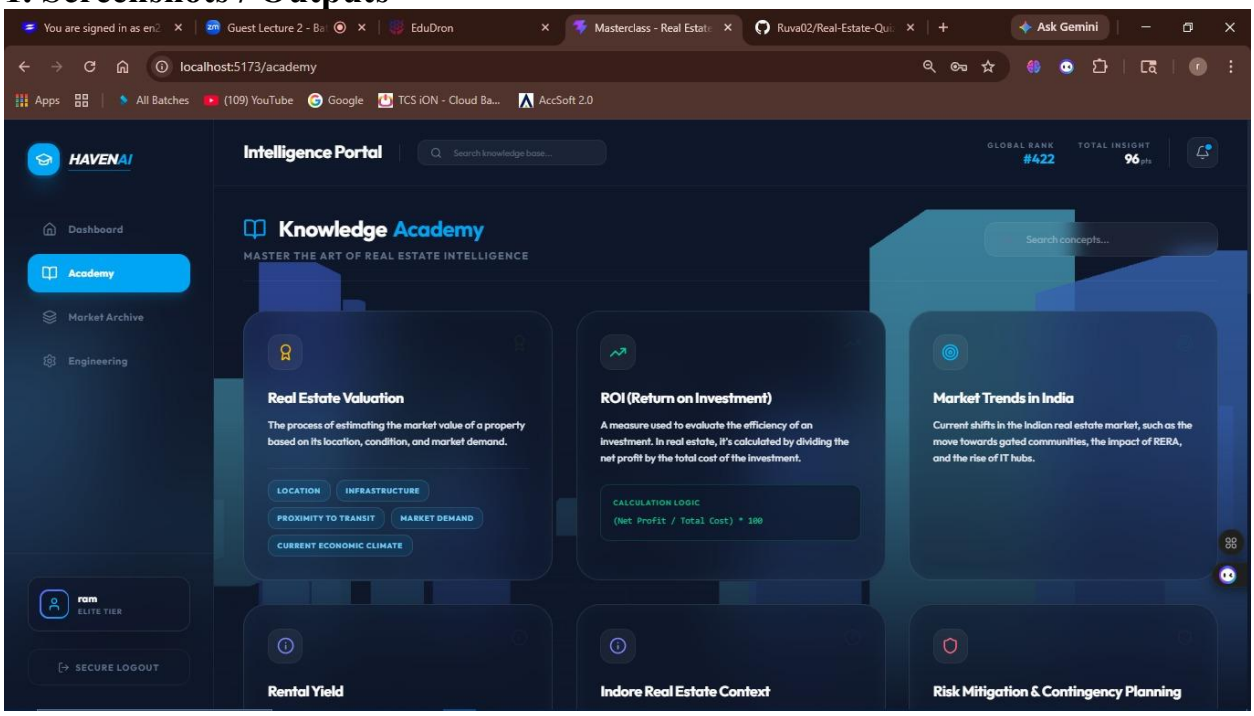
The frontend of the system is developed using **React**, a modern JavaScript library used for building dynamic and responsive user interfaces. React enables the creation of reusable components, which helps in managing the quiz interface efficiently. It allows users to interact with the Real Estate Tutor Bot by starting quizzes, viewing AI-generated questions, submitting answers, and receiving feedback in real time. The use of React ensures a smooth, fast, and user-friendly experience with efficient state management and seamless updates without reloading the page.

Backend

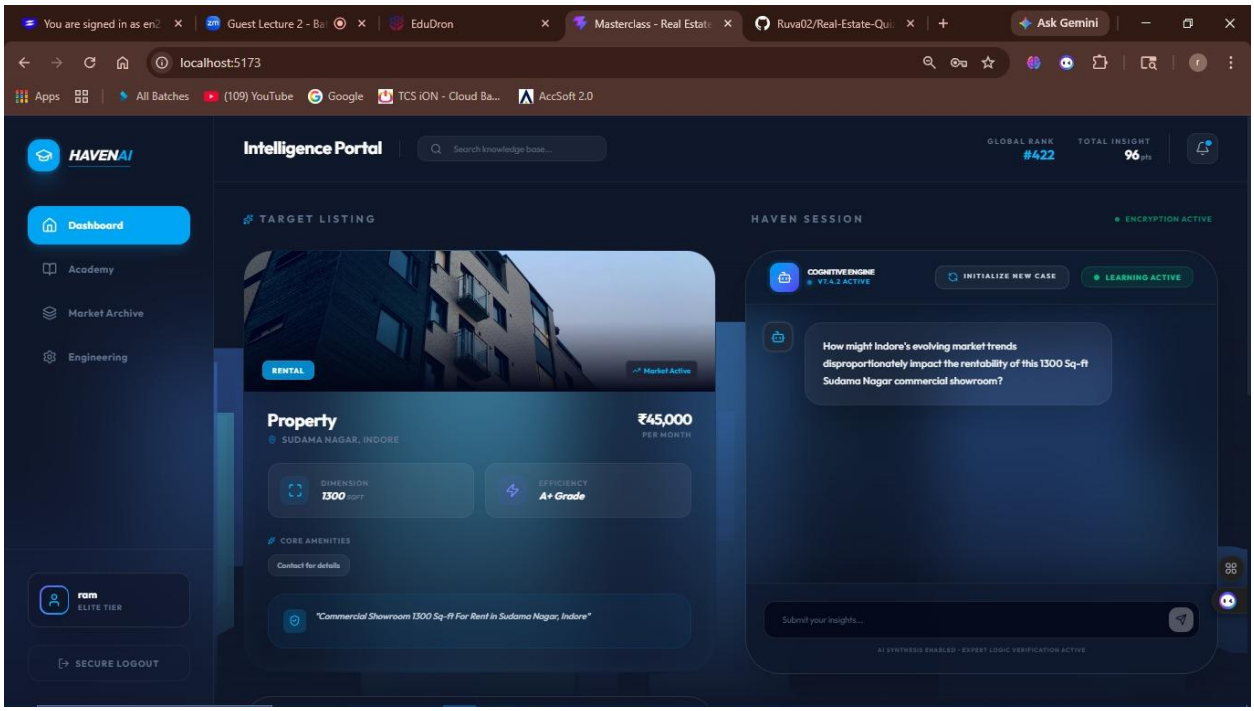
The backend of the system is implemented using **FastAPI**, a high-performance Python framework used for building APIs. FastAPI handles all the core logic of the application, including managing quiz sessions, processing user inputs, and communicating with the Generative AI module. It integrates with the Groq API to generate questions, evaluate answers, and provide explanations. The backend also manages local storage operations and ensures efficient communication between different system components. FastAPI is chosen for its speed, simplicity, and ability to handle asynchronous requests efficiently.

Results & Output

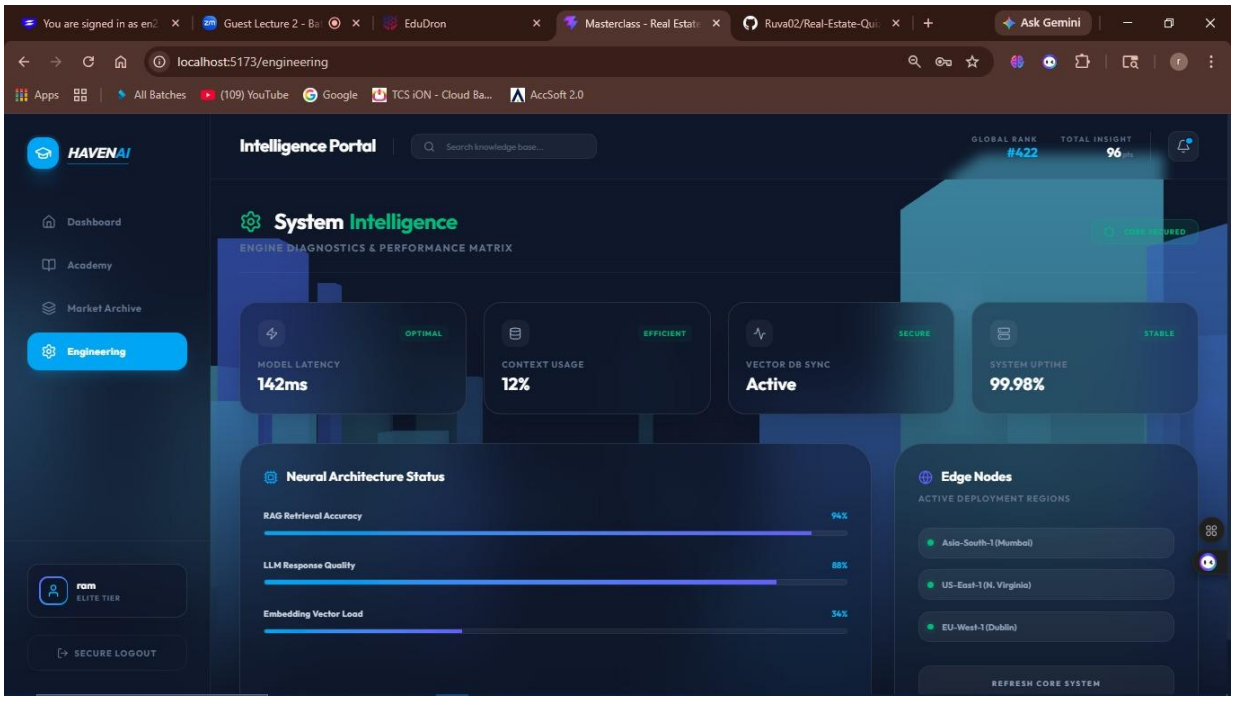
1. Screenshots / Outputs



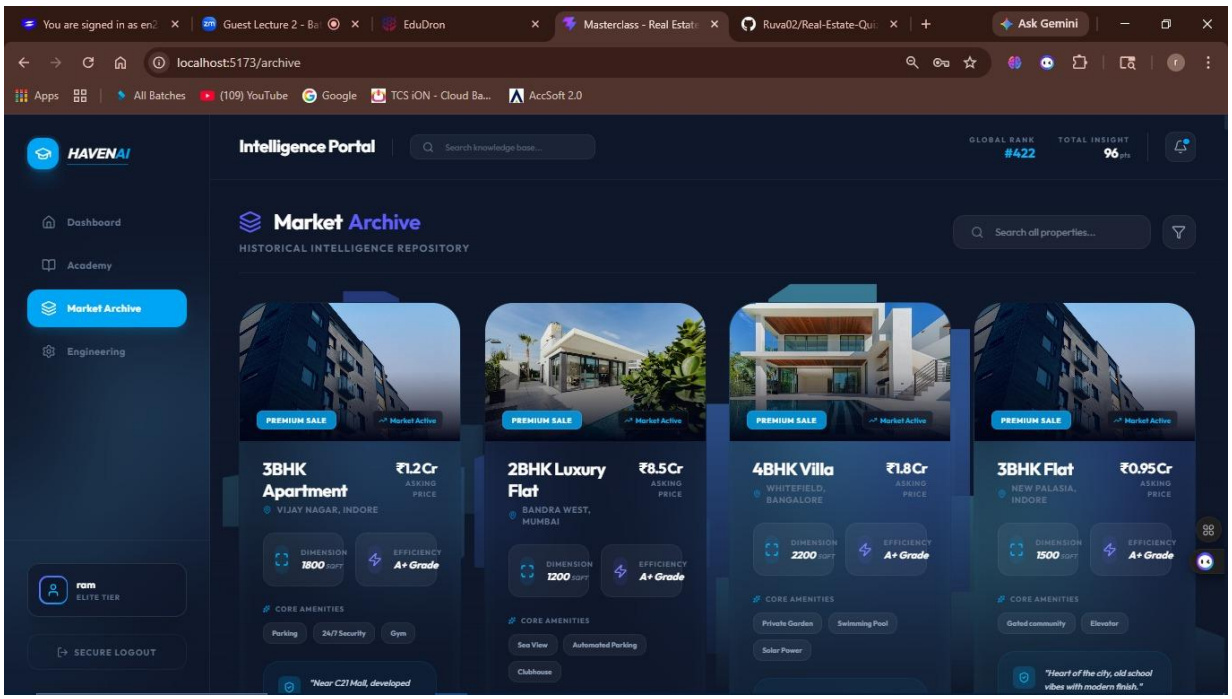
Fields Overview



DashBoard



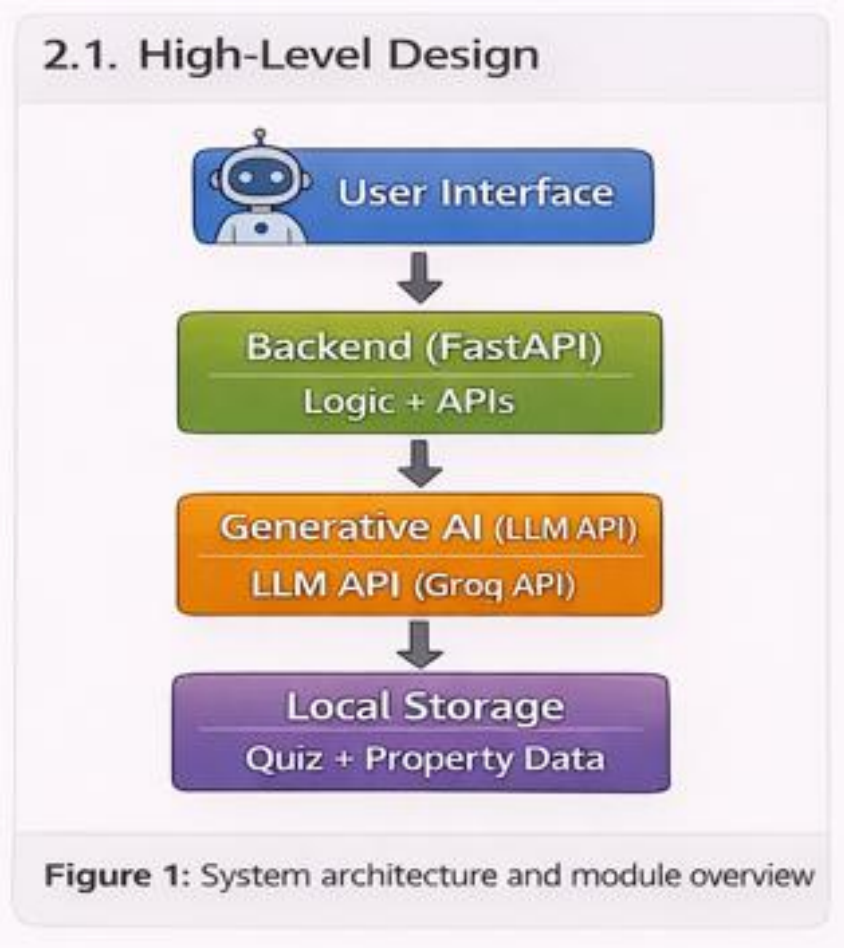
Personal Progress



Property Archive

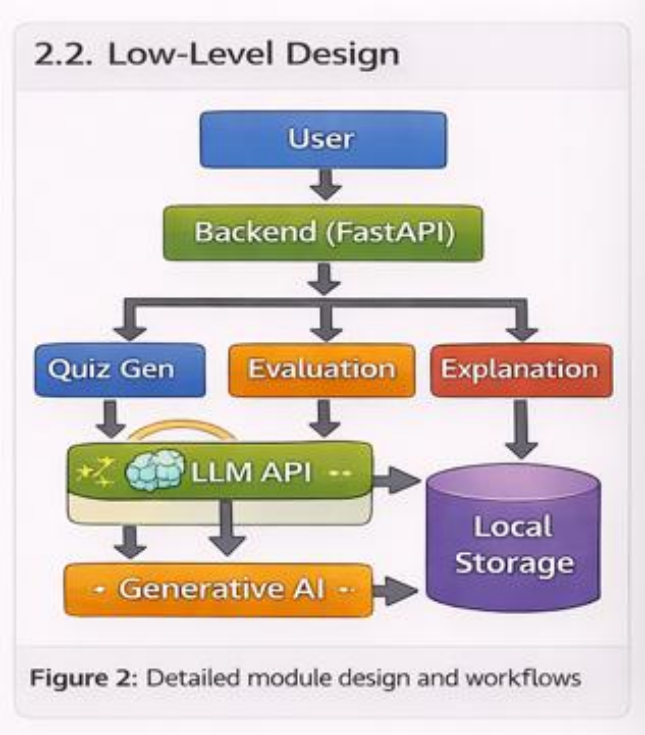
2. Reports / Dashboards / Models

2.1 High-Level Design (HLD)



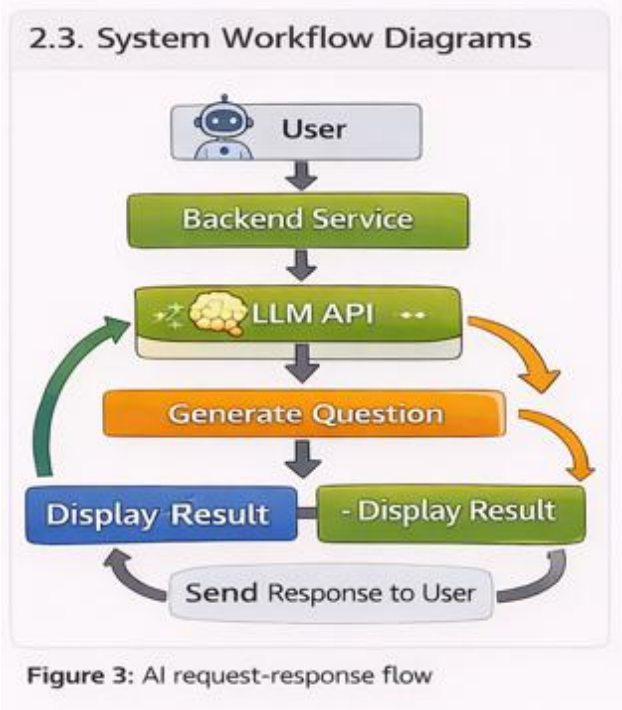
System architecture and module overview.

2.2 Low-Level Design (LLD)



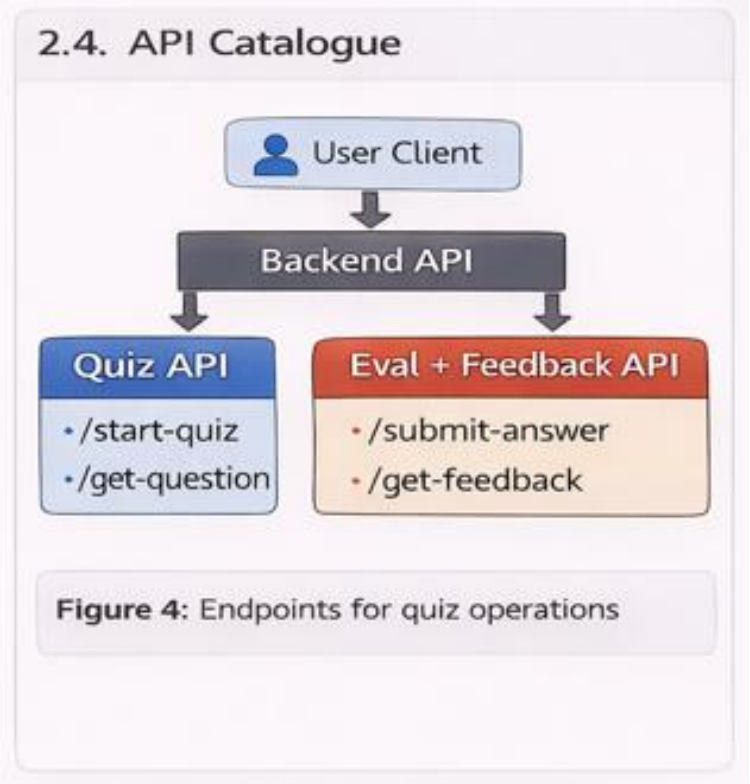
Detailed module design and workflows.

2.3 System Workflow Diagrams



AI request-response flow.

2.4 API Catalogue



Endpoints for quiz operations.

3. Key Outcomes

The project successfully demonstrates the use of Generative AI in the education domain.

Key outcomes include:

- Dynamic question generation
- semantic evaluation
- personalized explanation
- improved learning flow
- real estate domain specialization

The project highlights how AI can transform static quiz systems into intelligent tutoring platforms.

Conclusion

The **Real Estate Quiz / Tutor Bot** successfully demonstrates the practical application of Generative AI in the field of education and domain-specific learning. The system transforms a traditional quiz-based approach into an intelligent and interactive learning platform by leveraging the capabilities of Large Language Models (LLMs). Instead of relying on static questions and simple scoring, the system dynamically generates questions and evaluates user responses based on their contextual meaning, making the evaluation process more accurate and insightful.

One of the key achievements of this project is the ability to provide **personalized feedback** through AI-generated explanations. This not only helps users understand their mistakes but also strengthens their conceptual knowledge in the real estate domain. The continuous feedback loop created by the system enhances learning efficiency and keeps users engaged throughout the quiz process.

The integration of technologies such as React for the frontend, FastAPI for the backend, and Groq API for LLM processing ensures that the system is both **efficient and scalable**. The modular architecture allows each component to function independently while maintaining seamless communication, which makes the system easier to maintain and extend.

Additionally, the use of local storage simplifies data management while ensuring quick access to quiz-related information. The system is designed with performance and usability in mind, providing fast response times and a smooth user experience. Overall, the project highlights how Generative AI can be effectively utilized to create intelligent educational tools that go beyond traditional methods and offer a more engaging and adaptive learning environment.

Future Scope & Enhancements

The Real Estate Quiz / Tutor Bot has strong potential for further development and enhancement. While the current system provides a solid foundation using Generative AI, several advanced features can be integrated to improve its functionality, scalability, and user experience.

One of the major areas of enhancement is the integration of **cloud-based storage and deployment**. Moving from local storage to cloud databases such as MongoDB Atlas or Firebase would enable better scalability, real-time data access, and support for a larger number of users. Additionally, deploying the system on cloud platforms would make it accessible globally and improve system reliability.

Another important improvement is the implementation of **adaptive learning mechanisms**. The system can be enhanced to adjust the difficulty level of questions based on the user's performance, providing a more personalized learning experience. This would allow beginners and advanced users to benefit equally from the platform.

The addition of a **mobile application** can significantly increase accessibility and user engagement. Users would be able to access the tutor bot anytime and anywhere, making the learning process more flexible. Furthermore, incorporating **voice-based interaction** using speech recognition and text-to-speech technologies can make the system more interactive and user-friendly.

The system can also be extended to support **multilingual capabilities**, enabling users from different regions to interact with the bot in their preferred language. This would broaden the reach of the application and make it more inclusive.

Another valuable enhancement is the integration of **real-time property data APIs**, which would allow the system to generate questions based on current market trends and real-world data. This would make the learning experience more practical and relevant.

Advanced features such as **performance analytics dashboards** can also be added to track user progress over time. These dashboards can provide insights into strengths, weaknesses, and improvement areas, helping users monitor their learning journey effectively.

Finally, the system can be expanded beyond the real estate domain to include other educational fields such as finance, healthcare, or technology, making it a versatile AI-based tutoring platform.